

Barkot Tamrat Simegn

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EDUCATION

Northwestern University Evanston, IL	June 2027
Bachelor of Science in Mechanical Engineering GPA 3.3/4.0	
Community College of Aurora Aurora, CO	May 2023
A.S. in Physics GPA: 3.8/4.0	
Overland High School Aurora, CO GPA: 4.0/4.0	May 2023
Relevant Courses: Material Science, Mechanics of Materials, Thermodynamics, Differential Equations, Fluid Dynamics, Statics, Dynamics, Multivariable Calculus, Scientific and Embedded Programming in Python, Electronics Design, Linear Algebra	

SKILLS

Technical Skills: Prototyping, Python, Circuit Python, Matlab, Breadboarding, Microcontrollers, CAD, CAM, Certified Solidworks Professional, Certified Solidworks Associate, Microsoft Suite, Soldering, Excel, HTML/CSS, Spanish

PROJECTS

Northwestern Underwater Robotics Frame Team	Sep 2024 - Present
- Design and build a brand new frame design for an underwater robot using Solidworks and Onshape. - Manufactured using a variety of methods including water jetting HPV, milling brackets, and 3d printing fan covers. - Assembled and tested an underwater robot frame and iterated upon the design for a competition.	
Modified Bike Stand Shirley Ryan Ability Lab	March - June 2024
- Collaborated with a multidisciplinary team to create a bike stand for a student with cerebral palsy, enabling independent use and increased accessibility. - Conducted research, user interviews, and data collection to inform design decisions to meet specific accessibility needs. - Utilized Solidworks for computer-aided design and 3D printing technology to create and test functional prototypes.	
Northwestern NSBE Finance Chair McCormick School of Engineering	Jan-April 2024
- Managed finances for the Northwestern University chapter of the National Society of Black Engineers (NSBE), overseeing fundraising initiatives and chapter events. - Coordinated chapter meetings and events, ensuring effective communication and collaboration among team members. - Gained leadership experience through event management, financial oversight, and community engagement.	
Ostomy Bag Cleaner Shirley Ryan Ability Lab	Sep - Dec 2023
- Designed a compassionate and functional solution for cleaning ostomy bags, improving user independence and quality of life for patients. - Conducted primary research, including interviews with patients and medical professionals, to understand user needs and design requirements. - Utilized Solidworks for prototyping and 3D printing to develop and test a working model, culminating in a viable product ready for implementation.	

PROFESSIONAL EXPERIENCE

Manager City of Aurora Aurora CO	May 2022 - Present
- Supervised daily operations at various city facilities, ensuring patron safety, facility maintenance, and operational efficiency. - Directed many different teams of staff responsible for maintaining facility cleanliness, executing routine safety inspections, and enforcing safety protocols. - Oversaw dozens of training sessions to ensure team members were up-to-date on safety regulations and operational procedures. - Promoted from a temporary position to Manager on Duty, demonstrating leadership, conflict resolution, and team management skills through consistent performance.	
Cardboard Carnival Y.O.U. Maker Space Evanston IL	Jan-April 2024
- Orchestrated educational workshops for underrepresented youth, introducing foundational concepts in engineering, design, and coding. - Taught a group of 20 students how to write basic code to control servo motors, fostering problem-solving skills and creativity through hands-on projects. - Mentored students in collaborative design thinking, encouraging teamwork and iterative development processes. - Gave back to the community by inspiring and empowering youth to explore STEM.	